Project Ditto v3 — Pre-Registration Specification

Specification v1.0 · Pre-registration draft Full PDF: `SPEC.pdf` (immutable — do not edit after freeze)

> **This document will be frozen before evaluation begins.** Thresholds, methodology, and analysis plan are committed here before T-code implementation begins. Any methodology change after Gate 1 invalidates the pre-registration and requires a new dated spec.

Hypothesis (precisely scoped)

The constraint-chain abstraction developed in Project Ditto (six constraint types: `ResourceBudget`, `ToolAvailability`, `SubGoalTransition`, `InformationState`, `CoordinationDependency`, `OptimizationCriterion`) captures generalizable structure in sequential decision-making, AND **training-data exposure compresses the magnitude of the real-vs-shuffled detectability gap**. When applied to formal-rule game-playing trajectories from two game families (chess and checkers), each with a high-exposure standard variant and a low-exposure structural variant (Chess960 and international draughts), the abstraction will produce a stronger detectability gap in the lower-exposure variant of each family.

The experiment is a **directional hypothesis test of training-exposure as a gap-compression mechanism**, conducted within a generality-test framing. It extends v1 (Pokémon) and v2 (programming) by adding a third domain family (formal games) and by directly testing one mechanistic explanation for v2's observed Sonnet-vs-Haiku gap compression.

What this experiment is NOT

- Not a test of whether game telemetry helps programming reasoning
- Not a cross-domain transfer test
- Not a fine-tuning experiment
- No Pokémon-derived or programming-derived data enters this evaluation
- Does not measure game-playing capability
- Does not justify any production application
- Does not test all confounds simultaneously — chess vs. Chess960 isolates exposure-within-structure; checkers vs. draughts replicates that isolation in a second game family

Pre-registered Success Criteria

The criteria below test a *pattern* of per-cell results across the four primary cells, not a single contrast statistic. Per-cell tests are well- powered under the v3 sample plan; the contrast itself is reported as a descriptive supporting analysis but is not the hypothesis test.

Per-cell thresholds (each cell tested independently)

Criterion	Threshold	Tier
Layer 1 actionable gap (real − shuffled, both-actionable filter)	≥ 0.05	Moderate-positive
Layer 1 actionable significance	Bonferroni-corrected $p < 0.05$	Moderate-positive

Layer 2 gap (legality × optimality composite)	≥ 0.04	Layer 2 confirmation
Layer 2 significance	Bonferroni-corrected $p < 0.05$	Layer 2 confirmation
Strong-positive	Layer 1 actionable gap $\geq 0.08 \wedge$ Bonferroni $p < 0.01$	Strong-positive

Pre-registered hypothesis-pattern outcomes

The hypothesis is supported, partially supported, refuted, or inconclusive based on the joint pattern across the four primary cells.

Hypothesis supported (training-exposure compresses gap): Both of the following hold:

- (Chess family) Standard chess clears at most moderate-positive AND Chess960 clears strong-positive
- (Checkers family) American checkers clears at most moderate-positive AND international draughts clears strong-positive

Hypothesis partially supported: The pattern holds in exactly one game family but not the other, OR the direction is consistent across both families but neither low-exposure variant clears strong-positive while both clear moderate-positive at higher magnitude than their high-exposure counterparts (gap difference ≥ 0.03 in the predicted direction within each family).

Hypothesis refuted: Any of the following:

- Both standard chess and Chess960 clear strong-positive (no compression visible despite large exposure differential)
- Both American checkers and international draughts clear strong-positive (same; second family)
- Low-exposure variants show *smaller* gaps than their high-exposure counterparts in both families (wrong direction)

Inconclusive:

- Any cell fails to reach the per-cell sample-size target due to data acquisition or chain-construction issues
- Per-cell power post-hoc falls below 70% for the relevant threshold
- Three or more cells return null results, suggesting chain-construction failure rather than informative absence of effect

Minimum publishable result (independent of hypothesis outcome)

At least one cell clears the moderate-positive threshold at Bonferroni- corrected significance under the pre-registered methodology. This is the v2-style minimum and is met independently of the hypothesis-pattern outcome.

Outcome tiers (consistent with v1 and v2): `strong_positive`, `moderate_positive`, `weak_mixed`, `null`, `reversed`.

Phase Structure (sequential / adaptive design — pre-registered)

v3 runs in two phases with a pre-committed gating rule between them.

Phase 1: Haiku evaluation (4 cells)

Run all four primary cells on Claude Haiku 4.5 at full pre-registered sample size before any gap statistics are

computed across cells. Technical monitoring (API errors, malformed chains, missing pairs) is permitted during the run; effect-size monitoring is not.

Phase 1 → Phase 2 gate (pre-committed before any data collection)

Run Sonnet (Phase 2) iff at least one of:

1. Any Haiku cell clears moderate-positive at Bonferroni-corrected $p < 0.05$ under the Haiku-only divisor of 4 (gap ≥ 0.05).
2. The within-family gap difference in either chess or checkers reaches ≥ 0.03 in the predicted direction (Chess960 > standard chess, OR draughts > American checkers), regardless of per-cell significance.
3. Haiku per-cell results are mixed in a way Sonnet would resolve: exactly one cell clears moderate-positive AND its within-family partner falls between gap = 0.02 and gap = 0.05 (boundary case).

Skip Sonnet (Phase 1 only is the final result) iff all of:

1. All four Haiku cells have gaps within 0.015 of each other.
2. No Haiku cell reaches Bonferroni-corrected $p < 0.05$ (divisor 4).
3. No within-family gap difference exceeds 0.02 in either direction.

If neither the run nor the skip criteria are met (ambiguous Phase 1 result), default behavior is to **run Phase 2** to disambiguate. This is intentionally asymmetric — the cost of Phase 2 is bounded; the cost of an inconclusive v3 is much higher.

Bonferroni divisor

- If Sonnet runs: divisor = 8 (4 cells × 2 models)
- If Sonnet does not run: divisor = 4 (4 cells × 1 model)
- The divisor used in the final report is determined by which phases ran, not by post-hoc convenience.

Adaptive design disclosure

This sequential design is itself a pre-registered methodology choice. The gate criteria above are committed before Phase 1 begins. No effect-size peeking within Phase 1 is permitted. Phase 2 design is not modified based on Phase 1 results (only the decision to run or skip Phase 2 is adaptive).

Models and Evaluation Parameters

Parameter	Value
Models	Claude Haiku 4.5 (`claude-haiku-4-5-20251001`), Claude Sonnet 4.6 (`claude-sonnet-4-6`)
Primary config	Temperature 0.0, seed 42
Variance study	T = 0.5, seeds 1337 and 7919
Max tokens	50
Cutoff K	`len(constraints) // 2`
Prompt version	v3.0-game
API	Anthropic Messages Batches (50% cost reduction)
Action normalization	Full normalization (case, punctuation, whitespace, word order)

The three-config variance study (matching v1 and v2) is included to characterize per-cell stability. The primary config result is the pre-registered headline; T = 0.5 configs are reported as supplementary variance characterization, not as additional cells in the Bonferroni family.

Data Sources (committed — will not change after freeze)

Chess family (game family 1)

Variant	Exposure	Source	Target chains
Standard chess	High	`Lichess/standard-chess-games` (HuggingFace, Parquet, CC0)	1,200 real chains
Chess960	Low	`Lichess/chess960-chess-games` (HuggingFace, Parquet, CC0)	1,200 real chains

Rating filter applied at load time: `WhiteElo ≥ 1800 AND BlackElo ≥ 1800`. Filtering is performed via `polars` or `pandas` on the Parquet files, not in-stream, to ensure deterministic sampling.

Rationale: Both sources are from the same platform (Lichess) maintained as the official Lichess HuggingFace organization. This reduces platform-specific stylistic confounds. Both are CC0-licensed; both are auto-converted to Parquet for fast filtering. The rating floor (1800) ensures non-trivial play in both variants. Exposure differential is structural — the standard chess corpus is dramatically larger than Chess960 in publicly available training data (orders of magnitude difference in human-played game counts across all sources, not just Lichess).

Access pattern: stream directly from HuggingFace via the `datasets` library (`load_dataset("Lichess/standard-chess-games", streaming=True)` and likewise for Chess960). No manual download required. For deterministic sampling under the rating filter, materialize a filtered subset to local Parquet/JSONL once during data validation (Session 2), then operate on the materialized subset for chain construction. This avoids re-streaming the full dataset on every chain-generation run while keeping initial acquisition lightweight.

PGN parsing: `python-chess` library. Chess960 starting FEN is provided per game in the dataset; pass it to `chess.Board(fen=starting_fen, chess960=True)` during parsing.

Explicitly excluded: `laion/strategic_game_chess` and any other engine self-play chess dataset. v3's hypothesis test depends on training-corpus exposure that reflects human play; engine self-play games have different distributional properties and would confound the exposure-mechanism test.

Checkers family (game family 2)

Variant	Exposure	Source	Target chains
American checkers (8x8)	Higher	Open Checkers Archive (OCA 2.0) PDN files; ACF tournament archives	1,200 real chains
International draughts (10x10)	Lower	FMJD database PDN tournament archives	1,200 real chains

PDN parsing: `pydraughts` library (AttackingOrDefending package on PyPI). This library supports both American (8x8) and Standard / International (10x10) variants in the same interface via `Board(variant="american")` and `Board(variant="standard")`. Single library, two variants.

Access pattern: PDN sources are not on HuggingFace and require download during data validation (Session 3). Canonical sources: OCA 2.0 archive at `http://fierz.ch/download.php`, ACF tournament archives, FMJD tournament archives at `https://fmjd.org`. Materialize to local JSONL once and operate on the materialized files for chain construction.

Rationale: Same family (move-and-capture), structurally analogous, substantially different corpus sizes. International draughts is geographically concentrated (continental Europe, francophone Africa); its training-data corpus is substantially smaller than American checkers, but the games have richer tactical structure (10x10 board, longer sequences). This is a known asymmetry — corpus difference and structural difference are confounded in this contrast. The *within-chess* contrast (standard vs. Chess960) is the cleaner test; the checkers family is a secondary replication.

Note on checkers data volumes: the checkers PDN sources (OCA, FMJD, ACF archives) are smaller than the chess HuggingFace datasets (tens of thousands of games rather than millions). The 1,200 chains/cell target should still be reachable, but if pilot inspection in Session 6 shows actionability rates below the assumed 83% retention through the both- actionable filter, the checkers cells may be the binding constraint.

Per-cell sample target: 1,200 real chains, generating ~1,000 paired evaluations after both-actionable filter

Power justification (full calculation in §Sample Size & Power):

- 1,000 paired evaluations per cell at v2-calibrated discordant rate gives ~94% per-cell power to detect gap $= 0.06$ at $\alpha = 0.0125$ (Haiku-only) and ~90% power at $\alpha = 0.00625$ (full Bonferroni).
- Per-cell power is the primary statistical concern given the reframed hypothesis structure.
- The descriptive contrast tests (chess vs. Chess960, checkers vs. draughts) have lower power and are reported with confidence intervals rather than as hypothesis tests.

Chain Construction (chess and checkers T-codes, pre-frozen design)

Chain definition

A constraint chain is a sequence of constraints derived from a 15–25 move window of a game, anchored to game phase. Chains are extracted from full games (not puzzles or tactical positions) to preserve the sequential-decision structure that produced detectable gaps in v1 and v2.

Constraint type mappings (frozen at T-code-game-v1.0)

Type	Chess (and Chess960)	Checkers (and draughts)
`ResourceBudget`	Material count and tempo (moves remaining, time budget if PGN includes clocks)	Piece count and tempo
`ToolAvailability`	Legal move set; piece mobility (count of legal moves per piece)	Legal move set; capture availability
`SubGoalTransition`	Phase transitions (opening → middlegame → endgame); plan changes inferred from pawn structure shifts	Phase transitions (opening → midgame → endgame); king-promotion subgoal
`InformationState`	Trivially complete (perfect information) — documented as a known asymmetry; entered into chains as constant `complete` to preserve six-type structure	Trivially complete — same treatment
`CoordinationDependency`	Piece coordination (defended pieces, X-rays, batteries); pawn structure dependencies	Piece coordination (back-row defense, double-corner control)
`OptimizationCriterion`	Implicit evaluation function (material, position, king safety) inferred from move choice	Implicit evaluation function (material, position, mobility)

Known asymmetry: `InformationState` is non-actionable in formal games

Chess and checkers are perfect-information games; `InformationState` is constant across all chains. This means v3 effectively tests 5 of 6 constraint types in the actionable subset (vs. v1 and v2 which test all 6). This is documented as a known structural property of formal-game chains, not a methodology change. The both-actionable filter still applies on the remaining 5 types. The constraint enters chains as a constant `InformationState(state="complete")` to preserve chain length and the six-type structure for cross-experiment comparability.

Pilot validation before full chain generation

Before generating the full 1,200 chains per cell, a pilot of 50 chains per cell is generated and inspected for:

- Constraint type distribution (expected: ~30% each of ResourceBudget, ToolAvailability, SubGoalTransition, with smaller fractions of CoordinationDependency and OptimizationCriterion)
- Chain length within target range (15–25 events post-aggregation)
- Leakage check pass rate (must be 100%; chains containing chess- or checkers-specific vocabulary fail rendering)
- Renderer output spot-check by both authors

Pilot must pass before full generation. T-code freezes at tag `T-code-game-v1.0-frozen` only after pilot inspection.

Renderer leakage vocabulary

Domain-blind rendering uses an extended programming-vocabulary leakage list from v2 plus chess- and checkers-specific terms:

- Chess: pawn, knight, bishop, rook, queen, king, castle, check, mate, fork, pin, skewer, en-passant, file/rank labels, algebraic notation
- Checkers: jump, capture, king (in checkers sense), crown, double-corner, square numbers (1-32 / 1-50)

Any rendered chain containing leakage vocabulary fails rendering. The leakage check is enforced in code; callers cannot bypass.

Pre-registered Statistical Methodology

Layer 1 (primary) — paired McNemar's test with continuity correction

For each cell, paired binary outcomes (top-3 match, real vs. shuffled) are analyzed with continuity-corrected McNemar's test. Pairs are aligned by `(base_chain_id, model, eval_seed)`. Pairs missing either real or shuffled result are reported and excluded.

Gap is computed as `real_match_rate - shuffled_match_rate` over all pairs (not just discordant), matching v2's match-rate-difference estimand.

Layer 2 (secondary) — paired t-test

Continuous Layer 2 scores (legality × optimality composite) are analyzed with paired t-test (`scipy.stats.ttest_rel`) on aligned pairs.

Both-actionable filter (Layer 1 actionable subset)

A pair enters the Layer 1 actionable analysis only if both the real and shuffled chains have an actionable constraint type at the cutoff position. Actionable types in the formal-game setting: `ToolAvailability`,

`SubGoalTransition`, `CoordinationDependency`, `OptimizationCriterion` (four types; `InformationState` is excluded as non-actionable; matches v2's filter logic adapted to formal games).

This is the corrected estimand from v2's correction process (matches `scorer_corrected_v2.py` filter logic). Pairs where one chain has an actionable cutoff and the other does not are excluded from the actionable subset analysis.

Multiple-comparisons correction

Bonferroni correction is applied across the primary cells:

- Phase 1 + Phase 2 (Sonnet runs): divisor = 8
- Phase 1 only (Sonnet does not run): divisor = 4

Per-config and per-source breakdown analyses are reported as supplementary and use a separate Bonferroni divisor calculated post-hoc; these are not primary cells.

Pair alignment

All result files are indexed by `(base_chain_id, model, eval_seed)` before scoring. The pair structure is:

```
(base_chain_id, model, eval_seed) × (shuffled_chain = base_chain_id +
    "_shuffled_" + shuffle_seed)
```

Three shuffled variants per real chain (seeds 42, 1337, 7919); each real evaluation contributes three pairs (one per shuffled variant) under the same eval_seed. This 3× replication is by design (matches v2) and increases McNemar test power without inflating real_rate.

Sample Size & Power

Calibration

Power calculations are calibrated against v2 Haiku-TB (the cell that cleared the pre-registered threshold under corrected paired analysis):

- n_pairs = 312 (after both-actionable filter)
- gap = 0.0801
- discordant rate (b+c)/n ≈ 0.228
- p_b = b/(b+c) ≈ 0.676 at gap 0.08

These are the closest analog to expected v3 cell behavior (Haiku, paired McNemar with both-actionable filter, structurally similar chain length).

Per-cell power table (computed under v2 calibration)

At Haiku-only divisor (α = 0.0125):

n_pairs	gap = 0.04	gap = 0.06	gap = 0.08
500	0.27	0.63	0.91
800	0.45	0.87	0.99
1,000	**0.56**	**0.94**	**0.99**
1,500	0.78	0.99	1.00

At full Bonferroni divisor ($\alpha = 0.00625$, both phases):

n_pairs	gap = 0.04	gap = 0.06	gap = 0.08
500	0.19	0.54	0.86
800	0.36	0.81	0.99
1,000	**0.47**	**0.90**	**1.00**
1,500	0.70	0.99	1.00

Pre-registered target

1,000 paired evaluations per cell after both-actionable filter, requiring ~1,200 real chains per cell (assuming ~83% retention through the filter, calibrated against v2's actionable subset retention rate of ~40% for TB adjusted upward for formal-game expected actionability of ~60%; pilot will validate this estimate before full chain generation).

This target gives:

- ≥ 90% per-cell power for the moderate-positive threshold at full Bonferroni
- ≥ 99% per-cell power for the strong-positive threshold at full Bonferroni
- Modest contrast power (≈ 27% at the predicted 0.04 within-family difference) — descriptive only, not the hypothesis test

Power deficiency disclosure

The within-family contrast tests (chess vs. Chess960, checkers vs. draughts) are underpowered to detect a 0.04 gap difference at standard α (≈ 27% power at full Bonferroni). This is a deliberate design tradeoff: the per-cell pattern is the hypothesis test; the contrast statistics are descriptive. The contrast underpowering is disclosed in the spec and will be re-disclosed in the results write-up.

Pipeline Reuse from v1/v2

The following modules are inherited and not rewritten:

Module	Source	Modification
`src/filter.py`	v1/v2 (unchanged)	None — chain validity logic is domain-blind
`src/shuffler.py`	v1/v2 (unchanged)	None
`src/normalize.py`	v1/v2 (unchanged)	None
`src/scorer_corrected_v2.py`	v2	Adapted for v3 actionable types
`src/runner.py`	v2	Adapted for v3 sources
`src/reference.py`	v2	New active_pair logic (current_phase, last_move_type)

New modules required for v3:

- `src/translation.py` — game-domain T-code (replaces v2's programming T-code)
- `src/aggregation.py` — game event compression
- `src/parser_chess.py` — PGN parser via `python-chess` → TrajectoryLog
- `src/parser_checkers.py` — PDN parser via `pydraughts` → TrajectoryLog
- `src/renderer.py` — adapted leakage check with chess + checkers vocabularies

- `src/observability.py` — adapted entity labels for game domain

Pre-registered Supplementary Analyses

1. **Per-config variance study** — three temperature/seed configs (T = 0.0/seed = 42, T = 0.5/seed = 1337, T = 0.5/seed = 7919), reported as supplementary stability characterization for each cell.
2. **Within-family contrast tests** — chess vs. Chess960 and checkers vs. draughts gap differences with 95% confidence intervals. Descriptive only; not in primary Bonferroni family.
3. **Cross-family contrast** — chess family vs. checkers family pattern comparison.
4. **Comparison to v1 and v2 effect sizes** — v1 Pokémon-Sonnet gap (0.206), Pokémon-Haiku gap (0.066), v2 Haiku-TB actionable gap (0.0801). v3 results placed in this context.
5. **Constraint-type carrier analysis** — which constraint types carry the actionable signal in each cell, following v2's TB-vs-SWE carrier asymmetry methodology.
6. **Sonnet-vs-Haiku gap compression** — within each cell, compare Sonnet gap to Haiku gap to test whether the v2-observed gap compression replicates in the game domain.

What it would establish if the hypothesis is supported

> "The Project Ditto methodology measures more than the presence of abstract sequential structure — it also measures the degree to which a model has internalized that structure as surface pattern-matching versus deeper causal-structure detection. Training-data exposure compresses the real-vs-shuffled gap; this is empirically demonstrated in two structurally distinct game families (chess and checkers) using within-family controls (Chess960, international draughts) that hold game structure approximately constant while varying corpus size."

This would reframe the methodology from "detection tool" to "measurement tool" and would be a substantially stronger contribution than v2's generality result alone.

What it would establish if the hypothesis is refuted

> "The constraint-chain abstraction generalizes to a third domain family (formal-rule games), strengthening the v1 + v2 generality claim. Training-data exposure does NOT drive the real-vs-shuffled gap magnitude — the v2-observed Sonnet-vs-Haiku gap compression and any cross-domain gap variation must have a non-exposure cause. v4 will investigate alternative mechanisms."

This is a less exciting outcome but still a publishable result with a clean negative finding on a specific confound. The detection paper becomes stronger by virtue of having ruled out the exposure explanation. v4 directions are pre-specified in `PROGRAM_OUTLOOK.md`.

Authors and Review

- Lead: Safiq Sindha (Microsoft Azure Hardware PM, independent research)
- Co-author: [Co-author name] (Columbia University, systems engineering; Boeing FT) — reviews all major decisions; on all v3 publications
- Spec drafts circulated to co-author before commit
- Pre-registration commit signed off by both authors before evaluation begins

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Pre-registration v1.0 — drafted [DATE]. Prior work: Project Ditto v1 (Pokémon), Project Ditto v2 (programming).